

~~When Simpler Models Win: A Calibration Benchmark
of scGPT for Cell Type Annotation~~

**An Accuracy and Calibration Benchmark of scGPT Pipelines and
Matched Classical Models for Cell Type Annotation**

Alireza Khosravi^{1,2,3}, Arshia Khosravi³, Kamil Łangowski¹, Michał Sieczczyński^{1,5},
Krzysztof Pastuszak^{1,2,4}, Anna Supernat^{1,2,*}, Anna J. Żaczek^{1,2}

¹Laboratory of Translational Oncology, Medical University of Gdańsk, Gdańsk, Poland

²Centre of Biostatistics and Bioinformatics Analysis, Medical University of Gdańsk, Gdańsk, Poland

³Scientific Circle of Artificial Intelligence in Medicine, Medical University of Gdańsk, Gdańsk, Poland

⁴Department of Algorithms and System Modeling, Gdańsk University of Technology, Gdańsk, Poland

⁵Department of Software Engineering, Gdańsk University of Technology, Gdańsk, Poland

*Correspondence: Anna Supernat, abednarz@gumed.edu.pl

Abstract

Background. Single-cell large language models (scLLMs) such as scGPT promise transferable representations for cell-type annotation in single-cell RNA sequencing (scRNA-seq), but their statistical calibration under large-scale supervised within-atlas evaluation is largely unexamined. Poorly calibrated confidence scores can mislead downstream biological interpretation, particularly in clinical contexts.

Results. Across six public single-cell atlases (1.3 million cells; triple-negative breast cancer (TNBC), Indonesian peripheral blood mononuclear cells (PBMC), brain, multi-tissue tumour microenvironment, human and pig pancreas) we evaluated frozen scGPT probes (a linear probe and stronger non-linear probes, XGBoost and Random Forest), matched classical baselines (Logistic Regression, XGBoost, Random Forest trained on highly variable genes restricted to the genes present in the scGPT vocabulary), and one end-to-end fine-tuned scGPT experiment on TNBC. Accuracy was reported as Macro F1 with bootstrap 95% confidence intervals; calibration as the Expected Calibration Error (ECE). Classical Logistic Regression on the matched gene set reached Macro F1 = 0.94–0.99 and $ECE \leq 0.013$ on every atlas, matching or beating every scGPT pipeline. End-to-end fine-tuning of scGPT on TNBC reached Macro F1 = 0.975 / ECE = 0.003, with bootstrap confidence intervals overlapping those of scGPT + XGBoost (0.972) and below the matched-gene Logistic Regression (0.993).

Conclusions. Classical machine learning matched or outperformed every frozen scGPT pipeline we tested on accuracy, and the matched-gene Logistic Regression baseline also exceeded the TNBC end-to-end fine-tuning experiment. On calibration, the matched-gene classical baselines were well-calibrated on every atlas, whereas the standard scGPT-embedding+LogReg pipeline was substantially miscalibrated. Calibration metrics should be mandatory in scLLM benchmarking, and classical baselines should be evaluated under matched downstream learners and matched gene sets before any claim of scLLM superiority.

Foundation-model embeddings are used for single-cell annotation, but their value depends on biological holdout, comparator design and downstream learning. We compared frozen scGPT 0.2.5 embeddings with matched expression; both were paired with independently tuned Logistic Regression, Random Forest and XGBoost. Training-fitted SVD-512 was evaluated with Logistic

Regression across all atlases and with all classifiers only in TNBC and brain. Five donor-held-out folds yielded 965,427 analysed cells from 312 donors. Matched expression achieved higher pooled Macro F1 in six TNBC/PBMC comparisons; scGPT was higher in three brain comparisons. Donor-cluster bootstraps supported the six expression advantages and the brain Random Forest scGPT advantage, while the other brain intervals crossed zero. Where available, SVD-512 remained close to expression, arguing against compression alone as the explanation. Under strict PBMC label scarcity, expression reached Macro F1 0.818–0.930 versus 0.779–0.881 for scGPT. Held-out-donor temperature scaling reduced mean expected calibration error from 0.0072 to 0.0057 but not in every pipeline. Class analyses identified lymphocyte-boundary errors in TNBC and PBMC and rare-class errors in brain. Performance was atlas- and classifier-dependent for this frozen-scGPT pipeline; official fine-tuning, newer checkpoints and cross-cohort transfer were not tested.

Keywords: single-cell RNA sequencing; scGPT; cell-type annotation; calibration; donor-held-out validation; classical machine learning; foundation models

Introduction

Single-cell RNA sequencing (scRNA-seq) has made it possible to resolve heterogeneous tissues at cellular scale and to construct reference atlases spanning organs, developmental stages and disease states [1–6]. Interpreting these data still depends on cell-type annotation: the assignment of biologically meaningful identities to measured expression profiles. Manual annotation based on canonical markers remains valuable, but it is laborious, sensitive to expert judgement and difficult to scale. Automated methods therefore occupy a central place in atlas construction and reference mapping [7–12].

The available annotation strategies span a wide range of statistical complexity. Logistic regression, support-vector machines, nearest-neighbour projection, tree ensembles and reference-mapping systems can perform strongly when labelled cells from an appropriate reference population are available [7–9,13]. Latent-variable models such as scVI and scANVI additionally provide probabilistic representations and batch-aware integration [14,15]. These methods establish demanding baselines: a more complex representation is useful for supervised annotation only if it preserves task-relevant biological signal, transfers across the intended biological partition and improves an endpoint that matters in practice.

Single-cell foundation models seek to learn reusable representations from very large transcriptomic corpora. scBERT, Geneformer, scGPT, scFoundation and Universal Cell Embeddings differ in tokenization, architecture and pretraining objective, but share the premise that large-scale pretraining can encode regularities that are useful across downstream tasks [16–20]. scGPT is a generative transformer pretrained on more than 33 million cells and has been applied to cell-type annotation, integration, perturbation response and multi-omic analysis [18]. Its released whole-human checkpoint and public embedding interface make it a reproducible case study for asking whether a fixed pretrained cell representation improves supervised annotation.

The answer depends on the evaluation regime. Frozen embeddings followed by a supervised classifier are neither zero-shot annotation nor end-to-end fine-tuning. They test whether a fixed representation makes target labels accessible to a downstream learner. Strong performance in this regime may be valuable, but it does not establish transfer to unlabelled cell types, external atlases or biological conditions absent from the target training data. Conversely, weak frozen-embedding performance does not rule out benefits from official fine-tuning, other checkpoints or other foundation models. Recent independent evaluations have similarly emphasized that

foundation-model conclusions depend on task definition, pretraining exposure, comparator choice and adaptation protocol [21,22].

Biological partitioning is especially important in single-cell benchmarks. A random cell split allows cells from the same donor, sample or processing batch to appear in both training and testing. Such a split measures interpolation among cells from an atlas, but can yield optimistic estimates of deployment to unseen biological units. Grouped validation instead keeps all cells from a donor together, preserving the hierarchy of the data and making the test question explicit [23,24]. It does not by itself create external-atlas validation—cohort-specific acquisition, curation and label definitions remain shared—but it removes direct donor overlap and provides multiple biological test partitions.

Comparator construction also changes the scientific question. scGPT returns a 512-dimensional, unit-normalized cell embedding, whereas a direct expression baseline can contain more than one thousand sparse, log-normalized gene values on a different numerical scale. Applying the same regularization setting to both representations need not be neutral. Hyperparameters should therefore be selected separately inside the training data for every representation–classifier pair. A dimensionality-matched expression control is also informative: if direct expression outperforms a 512-dimensional embedding, a 512-component expression projection can help distinguish a scGPT-specific effect from the generic information loss associated with compression.

Annotation systems are often used through their confidence scores as well as their discrete predictions. Accuracy and Macro F1 assess discrimination, whereas calibration asks whether predictive probabilities agree with observed frequencies [25–27]. These properties are distinct. Expected calibration error (ECE) summarizes a reliability diagram but depends on binning; the multiclass Brier score and negative log-likelihood are proper scoring rules that jointly reflect

probability quality and discrimination [28–30]. Moreover, probabilities in a frozen-embedding pipeline are generated by the downstream classifier. Calibration findings must consequently be attached to complete representation–classifier pipelines, not attributed to the encoder alone. Post-hoc temperature scaling provides a simple test of how much probability error can be corrected without changing the predicted class [25].

We conducted an accuracy and calibration benchmark of official frozen scGPT embeddings, matched normalized expression and a 512-component expression projection. Frozen scGPT and matched expression were each paired with Logistic Regression, Random Forest and XGBoost, with hyperparameters selected independently inside the training donors. The primary evaluation used five donor-held-out folds in three human atlases spanning breast cancer, peripheral blood and brain. SVD-512 was evaluated as a dimensionality control with Logistic Regression in all three atlases and with all three classifiers in the TNBC and brain atlases. We further examined class-specific errors, labelled-cell scarcity with labelled-only feature selection, held-out-donor temperature scaling in two fully audited atlases and a separately reported pig-pancreas stress test using the human checkpoint. This design addresses a specific question: under supervised annotation of broad portal labels, how do fixed scGPT 0.2.5 embeddings compare with equally trained classical representations when the evaluated cells come from donors not used to fit the classifier?

Methods

Study design and estimands

The core benchmark compared two representations under three supervised classifiers. The representations were (i) frozen 512-dimensional cell embeddings returned by the official scGPT

interface and (ii) normalized expression of exactly the genes supplied to scGPT. Each was paired with Logistic Regression, Random Forest and XGBoost. A 512-component truncated singular-value decomposition (SVD) of the matched expression matrix provided a secondary dimensionality control. SVD-512 was paired with Logistic Regression in all three atlases and with all three classifiers in TNBC and brain. Hyperparameters were selected separately for every evaluated representation–classifier pair using only cells from the outer training donors. The primary estimand was annotation performance on cells from donors excluded from model fitting but drawn from the same curated atlas. It was evaluated with five-fold stratified group cross-validation in three human datasets. The comparison was prespecified at matched classifier: matched expression versus scGPT embeddings with Logistic Regression, with Random Forest and with XGBoost. SVD-512 comparisons were secondary controls for dimensional compression. No classifier was selected from outer-test performance for primary inference. A porcine dataset analysed with the released human checkpoint was treated separately as a cross-species stress test and was not included in the human benchmark summary.

Public single-cell datasets

Four processed, public H5AD objects were downloaded from CZ CELL×GENE Discover [31]. The primary human benchmark comprised TNBC breast cancer (427,823 cells; 101 unique donor identifiers), Indonesia peripheral blood mononuclear cells (PBMC; 462,034 cells; 199 unique donor identifiers) [32], and a human brain atlas (75,583 cells; 12 unique donor identifiers) [33]. Together these objects contained 965,440 cells before shared-input quality control. Pig pancreas (22,056 cells) originated from a cross-species islet study [34]; it contained one value of donor_id and two values of sample, could not support donor-held-out evaluation and was therefore analysed only as a stress test.

The direct download UUIDs, matrix dimensions, label counts and metadata audit are supplied with the code. The curated portal field `cell_type` defined the target. Labels represented by five or fewer cells were excluded before splitting; no retained class was subsequently removed. The human datasets contained 7–13 retained labels. These are portal-level cell categories rather than a harmonized fine-state ontology, and their biological granularity differs among datasets.

The grouping variable for every human dataset was `donor_id`. It was complete in all three objects. The metadata audit found 101, 199 and 12 unique donor IDs in TNBC, Indonesia PBMC and brain, respectively. The Indonesia object contained 199 unique `sample_id` values in one-to-one correspondence with the 199 donor IDs and 27 processing batches; the biological donor was used as the grouping unit.

Table 1. Datasets and biological partitions used in the benchmark. Cell counts precede shared-input quality control. Pig pancreas is not part of the primary human donor-held-out analysis.

Dataset	Cells	Classes	Donors	Analysis role
TNBC breast cancer	427,823	7	101	Five-fold donor-held-out human benchmark
Indonesia PBMC	462,034	13	199	Five-fold donor-held-out benchmark and scarcity analysis
Brain atlas	75,583	10	12	Five-fold donor-held-out human benchmark
Pig pancreas	22,056	4	1	Random-cell cross-species stress test

Pig pancreas contained two sample values but only one donor identifier. The whole-human checkpoint was applied without orthology mapping.

Outer donor-held-out partitions

~~For each dataset, cell labels were encoded as integer category codes. A stratified 80/20 train-test split was performed using `train_test_split` from `scikit-learn` (Pedregosa et al. 2011) with a fixed random seed of 42, ensuring that the class distribution was preserved across splits and results were reproducible.~~

For each human dataset, all cells belonging to one donor were assigned to the same fold. We used StratifiedGroupKFold with five folds, shuffling and deterministic dataset-specific seeds. Because rare classes can be concentrated in a small number of donors, the implementation searched a prespecified sequence of up to 100 seeds and retained the first assignment for which every class occurred in both training and test cells of every fold. The chosen seed is recorded for each dataset. Assertions verified that training and test donor sets were disjoint, that all retained classes were present on both sides of each split, and that the five outer-test index sets formed a non-overlapping out-of-fold partition after shared-input quality control. The training sets overlap across folds, as is intrinsic to cross-validation; the five folds are not described as independent training samples.

Training-only feature selection and shared gene input

~~HVG selection was performed on the full dataset prior to the train–test split, separately for the scGPT branch (flavor="seurat_v3", applied to the raw count matrix) and for the classical ML branch (flavor="seurat", applied to the normalised log-transformed matrix). Each flavour is the standard recommendation for its respective downstream consumer: seurat_v3 expects raw counts and is the preprocessing recipe followed by the official scGPT tutorials, whereas seurat is the classical Scanpy default for log-normalised matrices. The two pipelines therefore operate on different top-1,200 HVG sets—each prepared in the manner intended for the corresponding model family. We discuss the implications of this design choice—and of performing HVG selection prior to the train–test split—in the Limitations section.~~

All feature operations were repeated inside each outer fold. Raw integer counts were materialized from adata.raw for the outer-training cells only. Genes with fewer than three total counts in that training partition were removed, after which the top 1,200 highly variable genes

(HVGs) were ranked with the Seurat v3 variance-stabilizing method implemented by Scanpy [35,36]. This step did not access the outer-test cells.

Gene identifiers were compared with the released scGPT vocabulary by exact symbol, followed by uppercase-symbol matching when exact matching failed. Genes still absent from the vocabulary were removed, and duplicate mapped symbols retained only the highest-ranked training HVG. The resulting list was the sole input gene set for all three representations in that fold. Thus, direct expression did not receive genes unavailable to scGPT. Cells with zero counts across every shared input gene were excluded symmetrically before representation construction; exclusions and their original indices were recorded.

Frozen scGPT cell embeddings

The benchmark used scGPT 0.2.5 from source commit cebd6fa and the released whole-human checkpoint [18]. The checkpoint files were pinned by SHA-256 digest: args.json, c18e075e018140cb8b2d9029387b9de26607a5ce6a8ccabd6ead70cd76b95d60; vocab.json, acca93d114ca62c3f0f50debbd23e8c87f0714f4737764454f6b2b13f2e8580f; and best_model.pt, 6cb5d451ab5c4b33eb673adbe4fddc61d2389df1b89b7651a9fe2e557572b922.

Cell embeddings were extracted with the public scgpt.tasks.embed_data function, using raw counts, the matched feature_name symbols, max_length=1200, a batch size of 256, the released value-binning behaviour, fast-transformer execution and CUDA. The model weights were frozen. The function returned one 512-dimensional vector per cell. Every output was required to be finite, have the expected shape and have unit Euclidean norm within a tolerance of 10^{-5} . A downstream classifier was then trained with the target-atlas labels; accordingly, this is a frozen-representation supervised evaluation, not zero-shot annotation and not official end-to-end fine-tuning.

Matched expression and the SVD-512 control

For matched expression, counts of the same training-selected, vocabulary-matched genes were library-size normalized to 10,000 counts per cell and transformed with $\log(1+x)$ using Scanpy. The transformation was applied identically to outer-training and outer-test cells after the gene list had been fixed. Sparse zeros were retained as measured expression zeros.

The dimensionality control used randomized TruncatedSVD on the matched-expression matrix. Exactly 512 components were fitted to at most 100,000 outer-training cells, selected with class-stratified subsampling when necessary. The algorithm used five power iterations and a recorded fold-specific seed. Outer-test cells did not contribute to the decomposition. Truncated SVD was used rather than centred principal-component analysis because the expression matrix was sparse; the explained-variance ratio is supplied for every fold. This control asks whether reducing matched expression from approximately 1,100 variables to the same nominal dimension as scGPT accounts for the observed representation difference. It is not a generative or batch-corrected baseline and should not be interpreted as scVI.

Inner partitions and representation-specific hyperparameter selection

Outer-training donors were divided into disjoint inner partitions for hyperparameter fitting, validation and probability calibration. We created up to five stratified group folds, selected one class-complete group fold for validation, one different class-complete group fold for calibration and used the remaining folds for candidate fitting. Assertions excluded donor overlap among these partitions. For computational tractability, the hyperparameter fitting and validation partitions were separately capped at 50,000 cells using class-stratified subsampling. The calibration partition was not used in hyperparameter selection.

Three classical ML classifiers were trained on the top 1,200 HVGs (Seurat method) extracted from normalised, log-transformed count matrices:

- **Logistic Regression (Raw)** (Pedregosa et al. 2011): Multinomial L2-regularised logistic regression with solver="lbfgs", penalty="l2", C=1.0, class_weight=None, multi_class="auto", and max_iter=1000; convergence at this iteration cap was confirmed by the absence of scikit-learn non-convergence warnings on all six atlases. To clarify a possible ambiguity: the label "Raw" refers to the input *feature space* being the raw HVG *matrix* (i.e. no PCA, embedding, or other dimensionality reduction); the matrix itself is normalised and log-transformed exactly as for the other classical learners (Section "Data Preprocessing" above). It is *not* fit on raw count values.

The primary selection criterion was validation Macro F1. Candidate grids were fixed before outer-test evaluation and searched independently for every evaluated representation-classifier combination:

Logistic Regression: $C \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100, 1000\}$, with either the native representation scale or featurewise standardization. Sparse expression was scaled without centring. Models used the L2-penalized lbfgs solver with at most 5,000 iterations.

Random Forest: 300 trees, max_features $\in \{\sqrt{p}, 0.25p\}$ and min_samples_leaf $\in \{1, 5\}$, with bootstrap sampling and the Gini criterion [37].

XGBoost: histogram tree building with (max_depth, learning_rate, n_estimators) equal to (3, 0.10, 300), (6, 0.10, 300), (6, 0.05, 500) or (9, 0.05, 400) [38]. Expression matrices were materialized for XGBoost so that an absent sparse entry was treated as a measured zero rather than as a missing value.

260 Ties were resolved by candidate order. Once parameters had been selected, the corresponding
261 classifier was refitted on all outer-training cells and evaluated once on the outer-test donors. The
262 deterministic search code, candidate grids, selected parameters retained by completed jobs and
263 timings are supplied. Primary comparisons keep the classifier fixed, eliminating outer-test
264 selection of a preferred head.

265 **Discrimination and probability metrics**

266 Macro F1 was the primary discrimination metric because it weights each portal cell category
267 equally. For K classes,

$$268 \quad \text{Macro F1} = (1/K) \sum_k [2 \times \text{precision}_k \times \text{recall}_k / (\text{precision}_k + \text{recall}_k)].$$

269 Accuracy and balanced accuracy were secondary measures. We reported outer-fold values and
270 pooled out-of-fold values. Pooled values concatenate every donor's prediction exactly once and
271 therefore need not equal the unweighted mean of the five fold values.

272 Calibration and probability quality were described with five complementary statistics. Top-label
273 equal-width ECE used ten bins over [0,1]:

$$274 \quad \text{ECE} = \sum^b (|B^b|/n) \times |\text{accuracy}(B^b) - \text{confidence}(B^b)|, \text{ for ten confidence bins.}$$

275 We additionally calculated ten-bin adaptive ECE, ten-bin classwise ECE, the multiclass Brier
276 score and multiclass negative log-likelihood (NLL) [26,28,29]. Because ECE depends on binning
277 and all five metrics can be influenced by discrimination, no single statistic was treated as a
278 complete description of uncertainty. Reliability data retain the number of observations, mean
279 confidence and empirical accuracy in every bin.

Group-cluster bootstrap comparisons

~~All Macro F1 values from the controls are reported with bootstrap 95% confidence intervals (1,000 resamples of the test set, percentile method); the results appear in Section .~~

~~For each dataset and pipeline, out-of-fold predictions were converted to one confusion matrix per donor. We drew 5,000 bootstrap replicates by sampling the observed donors with replacement and summing all cells belonging to every selected donor. The same donor multiplicities were used for both members of a paired comparison. Macro F1 was calculated from the resampled confusion matrices, and the 2.5th and 97.5th percentiles of the paired difference formed a 95% interval. The primary contrast was matched expression minus scGPT embeddings for each fixed classifier; matched expression versus SVD-512 and SVD-512 versus scGPT were secondary.~~

This cluster bootstrap respects the donor as the sampling unit, but it does not refit HVG selection, SVD, hyperparameter tuning or classifiers inside each bootstrap replicate. Its intervals are therefore conditional on the completed five-fold set of fitted pipelines. They quantify sensitivity to the empirical donor composition, not the full uncertainty of rebuilding the entire analysis in a new cohort.

Held-out-donor temperature scaling

~~Statistical calibration was assessed using the Expected Calibration Error (ECE) (Guo et al. 2017), defined as:~~

~~We additionally note that ECE is one of several standard summaries of probability calibration, and that it is a non-proper summary statistic rather than a scoring rule. Two complementary strictly proper scoring rules are also commonly reported: the Brier score (Guo et al. 2017), a quadratic rule equal to the mean squared error between the predicted probability vector and the~~

one-hot true label; and the *negative log-likelihood* (NLL), a logarithmic rule that penalises confident errors far more steeply than the Brier score and is therefore especially sensitive to extreme overconfidence. These two rules are complementary rather than equivalent—they weight the same errors differently—and a full calibration audit would report all three together. We report ECE as the primary metric because it is the metric currently in use in the scLLM literature (Guo et al. 2017; Kedzierska et al. 2025) and because it underwrites the bin-level reliability diagram interpretation that motivates our calibration analysis. Because per-cell predicted probability arrays were not retained from the original runs, the scalar Brier and NLL values, together with signed and class-wise ECE decompositions, are left to future work; we therefore restrict our claims to the *magnitude* of miscalibration captured by ECE and the reliability diagrams.

Scalar temperature scaling was evaluated for Logistic Regression and XGBoost in all three representations in the TNBC and brain atlases [25]. Hyperparameters were chosen on the inner fitting and validation groups. A calibration-evaluation model was then fitted on the union of those two groups, leaving the disjoint inner calibration donors untouched. For Logistic Regression, the decision function supplied logits. For XGBoost, the logarithm of clipped class probabilities was used as a logit vector, which preserves the original softmax probabilities. A positive scalar T minimized calibration-donor NLL:

$$\hat{T} = \arg \min_{t_0} [-\sum_i \log \text{softmax}(z_i/T)_{y_i}].$$

The fitted temperature was applied once to the outer-test logits. Positive scalar scaling cannot change the class with the maximum logit; code assertions verified identical predicted classes before and after scaling. These calibration rows arise from classifiers trained without the

calibration donors and are therefore not numerically identical to the final discrimination models refitted on all outer-training donors.

Strict labelled-cell scarcity sensitivity

The Indonesia PBMC cohort supported a controlled label-scarcity sensitivity because it combined 13 labels with 199 unique donor IDs. The same five outer donor folds were used. Within each outer-training partition, we sampled exactly 10, 50, 100, 250 or 500 labelled cells per class with two deterministic sampling seeds. Crucially, gene filtering, Seurat v3 HVG ranking and vocabulary matching were recomputed from these labelled cells alone. Unlabelled outer-training cells were not used for feature selection, embedding extraction, classifier fitting or parameter selection.

The labelled and outer-test cells were represented by frozen scGPT embeddings and matched normalized expression. Logistic Regression parameters were selected independently for each representation by three-fold stratified cross-validation within the labelled subset using the same C and scaling grid, then refitted to all labelled cells. Evaluation remained on donors excluded by the outer fold. Results were first averaged across the two label-sampling seeds within fold and then summarized across five outer folds. This experiment concerns supervised few-label adaptation to known broad classes; it does not test zero-shot annotation, unseen label categories or discovery of rare states.

Biological error analysis

Out-of-fold predictions were aggregated into per-class precision, recall, F1 and support for every complete human representation–classifier pipeline. Complete confusion matrices were retained as counts and row-normalized proportions. The main text emphasizes the weakest classes and

largest off-diagonal confusions in TNBC, PBMC and brain rather than relying only on pooled scores. Errors were interpreted against the supplied portal annotations, which were treated as reference labels but not as an error-free biological gold standard.

Pig-pancreas cross-species stress test

~~The scGPT vocabulary is human. For the Pig Pancreas atlas we deliberately did *not* perform ortholog mapping: porcine gene symbols were matched directly against the human vocabulary (with an uppercase fallback), and any symbol still absent was masked with the <pad> token. This direct matching choice is itself part of the intended out-of-distribution stress test—it measures what a practitioner obtains by applying the released human scGPT model to non-human data without curation. Consequently, the Pig Pancreas scGPT result should be read as a lower bound on cross-species transfer rather than as an optimised cross-species pipeline; a dedicated ortholog-mapped evaluation is left to future work.~~

Pig pancreas was excluded from the primary human summary because its metadata contained one donor identifier and because the available checkpoint was trained for human data. We retained an explicitly secondary stress test to document the behaviour of the released pipeline under direct application. Porcine feature symbols were matched to the human vocabulary by exact name and uppercase fallback; no orthology map, one-to-one orthologue filtering or species-specific checkpoint was used. The analysis used one stratified random 80/20 cell split (seed 42), training-only HVG selection, the same three representations and independent inner hyperparameter tuning. It therefore measures within-object interpolation after an intentionally uncured human-to-pig mapping, not a competitive assessment of cross-species annotation.

Software, reproducibility and computational audit

Analyses ran under Python 3.11.10 with scGPT 0.2.5, PyTorch 2.5.1 and CUDA 12.4, AnnData 0.12.3, Scanpy 1.11.5, NumPy 1.26.4, pandas 2.3.3, SciPy 1.13.1, scikit-learn 1.9.0 [39] and XGBoost 3.2.0 on NVIDIA H100 80-GB GPUs. Seeds were fixed at every splitting, subsampling, decomposition and fitting stage. Each prediction archive records the dataset, fold, original test indices, donor IDs, class order, true labels and full probability matrix. Final aggregation was performed locally from already completed prediction files and admitted a pipeline only when all five aligned outer folds were present. Executable code, environment records, checkpoint digests, grouping audit, selected genes, available tuning traces, fold outputs and final tables are included in the submission package and public repository.

Results

A donor-held-out benchmark across three human atlases

~~We benchmarked four approaches across six diverse single-cell atlases comprising 1.3 million cells. The results are summarised in Figure and Table . We emphasise that “scGPT” in the figures and Table refers throughout to the end-to-end pipeline of frozen scGPT embeddings followed by a logistic-regression head; the comparison should therefore be read as *scGPT-embedding+LogReg* versus the classical baselines. A note on protocol: Table reports the standard scGPT linear-probe protocol following the reference implementation of Cui et al. (2024), whereas the fairness-control results in Section (Table) re-fit a plain sklearn LogReg on the same frozen embeddings under identical preprocessing as the vocabulary-matched controls. The two protocols yield slightly different point estimates for the scGPT + LogReg pipeline on TNBC (0.783 vs. 0.715) and Pig Pancreas (0.231 vs. 0.270) but the same qualitative conclusion~~

~~—scGPT is dominated by the classical matched-vocabulary baseline—on every atlas. Table—
should therefore be read as the initial reference-protocol benchmark; the primary controlled-
comparison is the fairness-control analysis (Table), in which downstream learners and input-
gene sets are matched across all models.~~

The primary analysis comprised 965,440 cells from three human atlases and 312 donor
identifiers before shared-input quality control (Figure 1; Table 1). Five outer folds were
completed per dataset. The aligned out-of-fold set contained 965,427 cells: 427,813 from TNBC,
462,034 from Indonesia PBMC and 75,580 from brain. Thirteen outer-test cells had zero counts
across the fold-specific shared input genes and were excluded symmetrically before
representation construction. No donor occurred in both training and test cells within a fold, every
retained portal class was represented on both sides of every split, and the five test sets contained
each retained cell once. Fold-specific vocabulary matching retained 1,167–1,172 genes in
TNBC, 1,132–1,141 in PBMC and 930–953 in brain (Table 2).

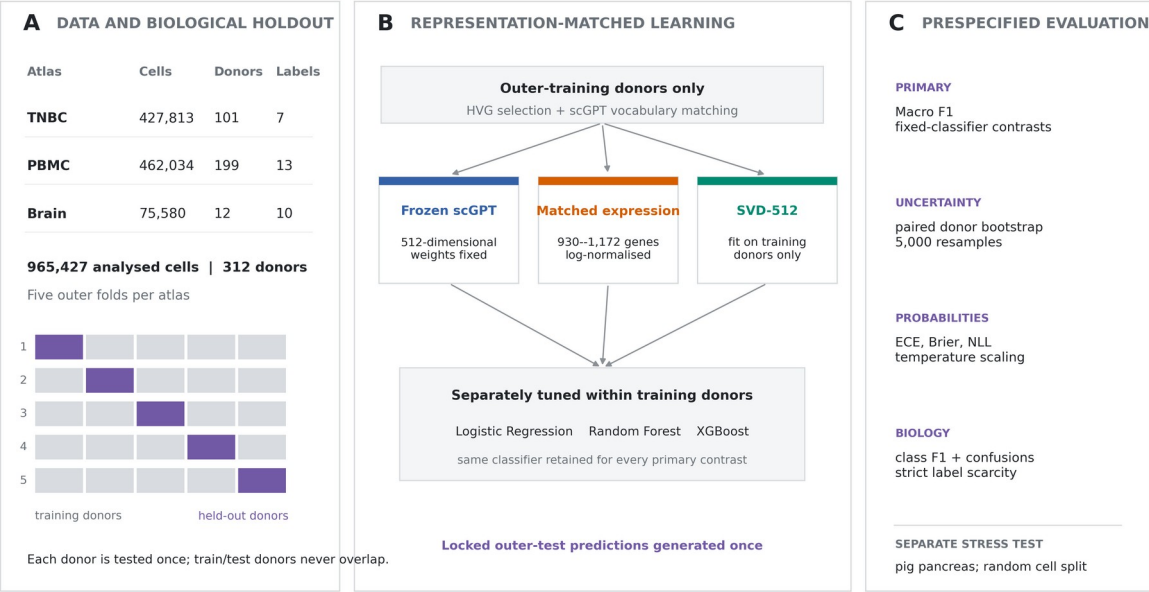


Figure 1. Study design. Three human atlases were evaluated with donor-held-out outer folds. Highly variable gene selection, vocabulary matching, SVD fitting, representation-specific hyperparameter selection and classifier fitting were restricted to outer-training donors. The primary contrasts compared frozen scGPT embeddings with matched expression at the same downstream classifier. Pig pancreas was excluded from the human summary and retained as a separate stress test.

Table 2. Completed out-of-fold design audit.

Dataset	Donor IDs	Classes	OOF cells	Matched HVGs
TNBC	101	7	427,813	1,167–1,172
PBMC	199	13	462,034	1,132–1,141
Brain	12	10	75,580	930–953

Representation effects depend on atlas and classifier

The performance gap between the scGPT pipeline and the classical methods is large, consistent, and dataset-invariant (Figure). Across all six atlases, Logistic Regression on raw HVGs achieved Macro F1 scores of 0.940–0.996, XGBoost achieved 0.932–0.997, and Random Forest achieved 0.883–0.995. In contrast, the scGPT linear-probe pipeline achieved Macro F1 scores of only 0.231–0.783—a gap of 0.21–0.76 F1 points relative to the best classical method on each dataset.

Matched expression had higher pooled out-of-fold Macro F1 than scGPT embeddings in six of the nine prespecified classifier-matched comparisons (Figure 2; Table 3). All six occurred in TNBC and PBMC. In TNBC, matched-expression Macro F1 was 0.990, 0.985 and 0.990 with Logistic Regression, Random Forest and XGBoost, compared with 0.983, 0.968 and 0.978 for scGPT embeddings. In PBMC, the corresponding values were 0.959, 0.933 and 0.957 for matched expression and 0.949, 0.915 and 0.935 for scGPT embeddings.

The ordering reversed in brain. scGPT embeddings achieved Macro F1 values of 0.940, 0.908 and 0.930 with Logistic Regression, Random Forest and XGBoost, whereas matched expression achieved 0.911, 0.870 and 0.925. The pooled brain accuracies were nevertheless 0.983–0.987, illustrating the sensitivity of Macro F1 to rare classes in this imbalanced atlas. No classifier was selected from these results: every value is one member of a prespecified fixed-classifier comparison. Complete accuracy, balanced accuracy, probability metrics and outer-fold results are reported in Supplementary Tables S3 and S4.

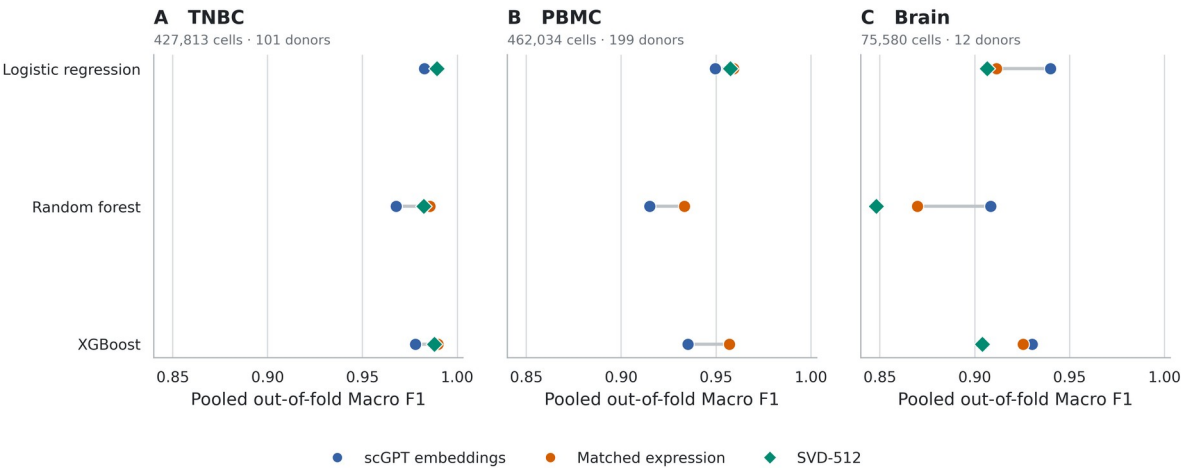


Figure 2. Donor-held-out annotation performance. Pooled out-of-fold Macro F1 for frozen scGPT embeddings, matched normalized expression and available SVD-512 controls. Frozen scGPT and matched expression were evaluated with all three classifiers in all three atlases. SVD-512 was evaluated with Logistic Regression in all three atlases and with all classifiers in TNBC and brain. A missing point denotes that a complete five-fold pipeline was not available and was not included.

Table 3. Primary classifier-matched donor-held-out performance. Values are pooled out-of-fold Macro F1; every retained cell contributes exactly once.

Dataset	Classifier	scGPT embeddings	Matched expression	SVD-512
TNBC	Logistic Regression	0.983	0.990	0.989
TNBC	Random Forest	0.968	0.985	0.982
TNBC	XGBoost	0.978	0.990	0.988
PBMC	Logistic Regression	0.949	0.959	0.958
PBMC	Random Forest	0.915	0.933	Unavailable
PBMC	XGBoost	0.935	0.957	Unavailable
Brain	Logistic Regression	0.940	0.911	0.906
Brain	Random Forest	0.908	0.870	0.848
Brain	XGBoost	0.930	0.925	0.904

PBMC SVD-512 Random Forest and XGBoost were unavailable as complete five-fold pipelines and were not imputed.

SVD-512 preserves most expression performance

The same-dimensional expression control remained close to matched expression. With Logistic Regression, SVD-512 reached Macro F1 values of 0.989 in TNBC, 0.958 in PBMC and 0.906 in brain, compared with 0.990, 0.959 and 0.911 for the uncompressed matched-expression input.

The corresponding expression-minus-SVD differences were 0.0005, 0.0016 and 0.0049. Donor-bootstrap intervals excluded zero for TNBC and PBMC but included zero for brain. Across the seven classifier-atlas comparisons for which both matched expression and SVD-512 had five complete folds, matched expression was higher in all seven; six conditional intervals excluded zero.

SVD-512 also preserved the atlas-specific ordering relative to scGPT. It exceeded scGPT with Logistic Regression in TNBC and PBMC, whereas scGPT exceeded SVD-512 in brain. Fold-specific SVD explained-variance ratios were 0.954–0.956 in TNBC, 0.958–0.959 in PBMC and 0.934–0.942 in brain (Supplementary Table S5); all SVD contrasts are reported in Supplementary Table S6. These results do not support dimensionality alone as an explanation for

the TNBC and PBMC differences, although they also show that a simple linear compression retained most of the matched-expression signal.

Donor-level intervals distinguish stable and uncertain orderings

~~The fairness-control protocol is described in full in Section . Briefly, on each of the six atlases we evaluated six models: scGPT embeddings paired with each of LogReg, XGBoost and Random Forest, and each of the three classical learners restricted to the vocabulary-matched HVG subset. All Macro F1 values are reported with bootstrap 95% confidence intervals (Table).~~

The paired donor-cluster bootstrap sampled all cells from a donor together and retained the pairing between representations (Figure 3; Table 4). In TNBC, the matched-expression advantages were 0.0071, 0.0177 and 0.0118 for Logistic Regression, Random Forest and XGBoost; the conditional 95% intervals were [0.0060, 0.0084], [0.0145, 0.0217] and [0.0099, 0.0141]. In PBMC, the differences were 0.0097 [0.0085, 0.0109], 0.0183 [0.0162, 0.0205] and 0.0217 [0.0199, 0.0236].

In brain, matched expression minus scGPT was -0.0285 [-0.0720, 0.0039] for Logistic Regression, -0.0387 [-0.0662, -0.0143] for Random Forest and -0.0048 [-0.0241, 0.0138] for XGBoost. Thus, the observed scGPT advantage was donor-bootstrap-secure for Random Forest but not for the other two classifiers. These intervals condition on the fitted out-of-fold pipelines; they do not incorporate rebuilding feature selection, decomposition, tuning and model fitting.

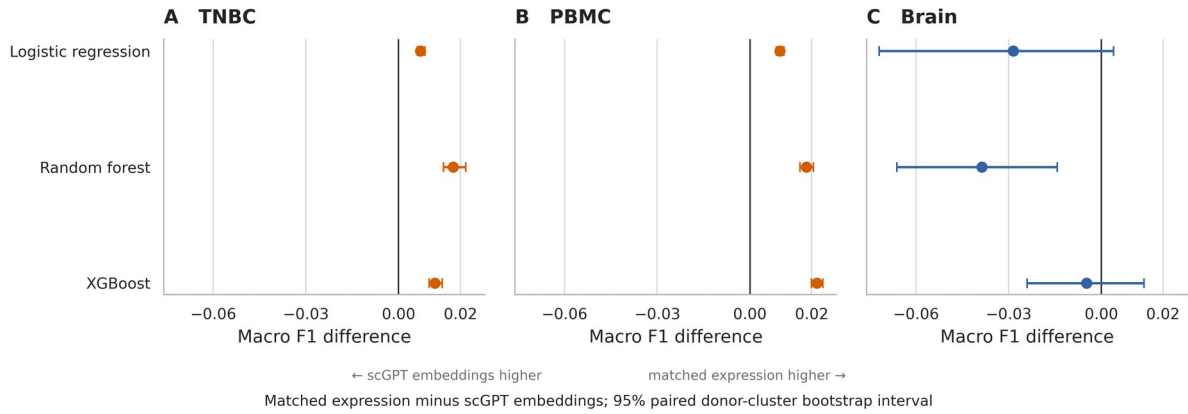


Figure 3. Paired donor-cluster bootstrap contrasts. Macro F1 difference between matched expression and frozen scGPT embeddings at a fixed classifier. Points are observed out-of-fold differences and bars are percentile intervals from 5,000 paired donor resamples. The resampling unit is the donor; fitted models are held fixed.

Table 4. Paired donor-cluster bootstrap contrasts for matched expression versus frozen scGPT embeddings. Intervals are conditional on the fitted out-of-fold pipelines.

Dataset	Classifier	Expression	scGPT	Difference	95% interval
TNBC	Logistic Regression	0.990	0.983	0.007	[0.006, 0.008]
TNBC	Random Forest	0.985	0.968	0.018	[0.014, 0.022]
TNBC	XGBoost	0.990	0.978	0.012	[0.010, 0.014]
PBMC	Logistic Regression	0.959	0.949	0.010	[0.009, 0.011]
PBMC	Random Forest	0.933	0.915	0.018	[0.016, 0.020]
PBMC	XGBoost	0.957	0.935	0.022	[0.020, 0.024]
Brain	Logistic Regression	0.911	0.940	-0.028	[-0.072, 0.004]
Brain	Random Forest	0.870	0.908	-0.039	[-0.066, -0.014]
Brain	XGBoost	0.925	0.930	-0.005	[-0.024, 0.014]

Calibration is a property of the complete pipeline

A key novel finding of our study is the systematic miscalibration of the scGPT-embedding+LogReg pipeline (Figure ; Table). Across all datasets, this pipeline exhibited ECE values of 0.038–0.177, indicating that its predicted confidence scores are substantially inconsistent with its actual accuracy. Specifically, the scGPT-embedding+LogReg pipeline is substantially miscalibrated: its predicted probabilities do not reliably track empirical accuracy. Because ECE is unsigned, we interpret the direction of this miscalibration only at the bin-level in

~~the reliability diagrams (Section). We note that the predicted probabilities are produced by the downstream logistic regression head fit on scGPT embeddings, not by scGPT itself; the calibration claim therefore concerns the end-to-end pipeline, not the foundation model in isolation.~~

Uncalibrated probability quality differed more strongly by classifier than by representation. Random Forest had the largest pooled equal-width ECE in TNBC and PBMC (0.037–0.059), whereas Logistic Regression and XGBoost were generally below 0.01. This pattern was visible for both scGPT embeddings and matched expression, arguing against assigning calibration behaviour to the encoder alone.

Held-out-donor temperature scaling was evaluated for Logistic Regression and XGBoost in TNBC and brain across all three representations (Figure 4; Table 5). Across the 60 outer-fold pipeline evaluations, mean equal-width ECE decreased from 0.0072 to 0.0057 and improved in 41 evaluations. Mean multiclass Brier score decreased from 0.02605 to 0.02591 and improved in 42 evaluations; mean NLL decreased from 0.0698 to 0.0610 and improved in 39. The remaining evaluations were unchanged or worsened, particularly where the unscaled predictions were already close to calibrated. Scalar temperature scaling preserved every predicted class, so discrimination was unchanged by construction. The Macro F1 values in the calibration supplement differ from the primary table because the calibration-evaluation classifiers deliberately exclude the inner calibration donors rather than being refitted on all outer-training donors.

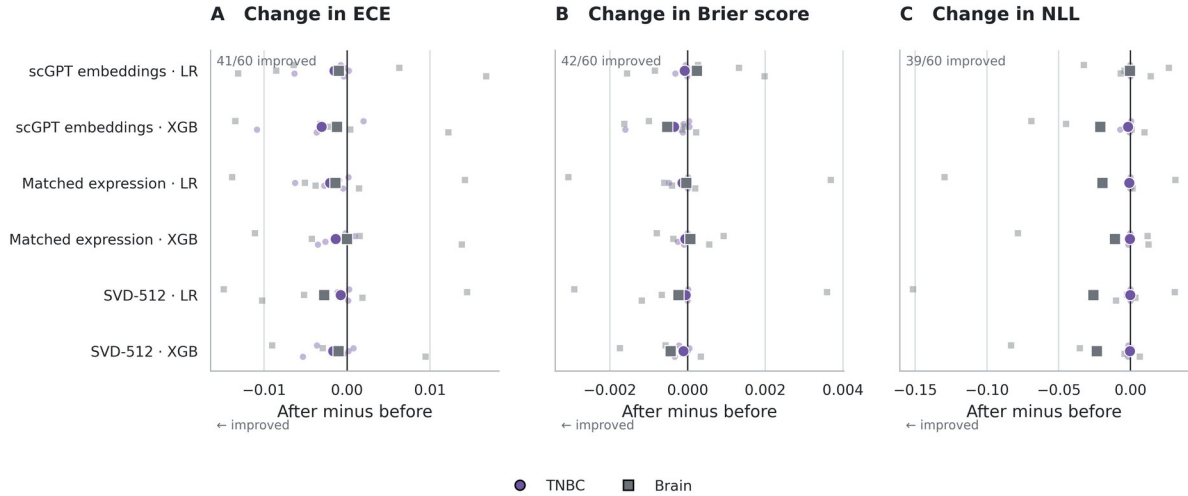


Figure 4. Held-out-donor temperature scaling. Fold-level changes after scalar temperature scaling for equal-width expected calibration error (ECE), multiclass Brier score and negative log-likelihood (NLL). Small points are individual outer folds; large points are dataset means. Negative differences indicate improvement. Circles denote TNBC and squares denote brain. LR, Logistic Regression; XGB, XGBoost.

Table 5. Held-out-donor temperature scaling, averaged over TNBC and brain (ten outer folds). Lower values are better.

Representation	Classifier	ECE before	ECE after	Brier before	Brier after	NLL before	NLL after
scGPT embeddings	Logistic Regression	0.0073	0.0060	0.0293	0.0294	0.0713	0.0708
scGPT embeddings	XGBoost	0.0083	0.0062	0.0337	0.0333	0.0873	0.0758
Matched expression	Logistic Regression	0.0069	0.0051	0.0231	0.0230	0.0630	0.0528
Matched expression	XGBoost	0.0069	0.0062	0.0221	0.0221	0.0602	0.0545
SVD-512	Logistic Regression	0.0066	0.0048	0.0239	0.0238	0.0673	0.0542
SVD-512	XGBoost	0.0070	0.0056	0.0242	0.0239	0.0696	0.0576

Matched expression retains an advantage under strict label scarcity

The Indonesia-PBMC sensitivity recomputed gene selection and vocabulary matching from the labelled subset alone. With 10 labelled cells per class, matched expression reached a mean donor-held-out Macro F1 of 0.818 (SD 0.025), compared with 0.779 (SD 0.032) for scGPT embeddings. At 50, 100, 250 and 500 labels per class, the corresponding values were 0.886 versus 0.841, 0.904 versus 0.855, 0.922 versus 0.872 and 0.930 versus 0.881 (Figure 5; Table 6). The fold-mean matched-expression difference ranged from 0.040 at 10 labels per class to

approximately 0.051 at 250 labels per class and was positive in every outer fold at every budget. This discrimination advantage did not imply uniformly better uncalibrated ECE: at 100–500 labels per class, the scGPT-embedding Logistic Regression pipelines had lower mean top-label ECE.

Table 6. Strict Indonesia-PBMC label scarcity. Feature selection and tuning used only the labelled cells. Macro F1 is mean $\pm$ SD across five donor-held-out folds after averaging the two sampling seeds within fold.

Labels per class	scGPT embeddings	Matched expression	Mean matched HVGs
10	0.779 $\pm$ 0.032	0.818 $\pm$ 0.025	1,169
50	0.841 $\pm$ 0.013	0.886 $\pm$ 0.019	1,160
100	0.855 $\pm$ 0.009	0.904 $\pm$ 0.013	1,156
250	0.872 $\pm$ 0.006	0.922 $\pm$ 0.006	1,154
500	0.881 $\pm$ 0.008	0.930 $\pm$ 0.008	1,149

Class-level errors reveal distinct biological regimes

The TNBC XGBoost pipelines performed strongly across all seven portal categories, but most remaining errors involved lymphocyte boundaries. scGPT embeddings assigned 9.9% of B-lineage lymphocytes to the T-cell class, compared with 3.7% for matched expression; the corresponding B-lineage F1 values were 0.916 and 0.967. Myeloid-to-T-cell confusion was 3.3% with scGPT and 1.5% with matched expression. These differences account for much of the aggregate TNBC advantage while showing that the broad stromal and endothelial categories were already near saturation.

PBMC contained more closely related immune categories. With XGBoost, scGPT embeddings confused 19.7% of naive CD8 T cells with CD4 T cells, whereas matched expression reduced this to 7.0%. Reciprocal CD8–natural-killer confusion remained prominent in both representations. The lowest scGPT class F1 was 0.830 for naive CD8 T cells, compared with 0.913 for matched expression. These errors are biologically plausible at broad portal boundaries.

536 | but also show that a pooled accuracy can conceal systematic redistribution among related
537 | lymphocyte states.

538 | Brain showed the opposite representation ordering and the strongest class imbalance. The rare
539 | central-nervous-system macrophage, perivascular-cell and pericyte categories each contained
540 | only 104–113 cells. Their XGBoost F1 values ranged from 0.775 to 0.911 with scGPT
541 | embeddings and from 0.777 to 0.853 with matched expression. Central-nervous-system
542 | macrophages were frequently assigned to microglia, while perivascular cells, pericytes and
543 | oligodendrocyte precursor cells were frequently assigned to oligodendrocytes. These rare-class
544 | errors explain why pooled accuracy exceeded 0.98 even when Macro F1 was substantially lower.
545 | Full class metrics and row-normalized confusion matrices are provided in Supplementary
546 | Figures S3–S4 and Tables S10–S11.

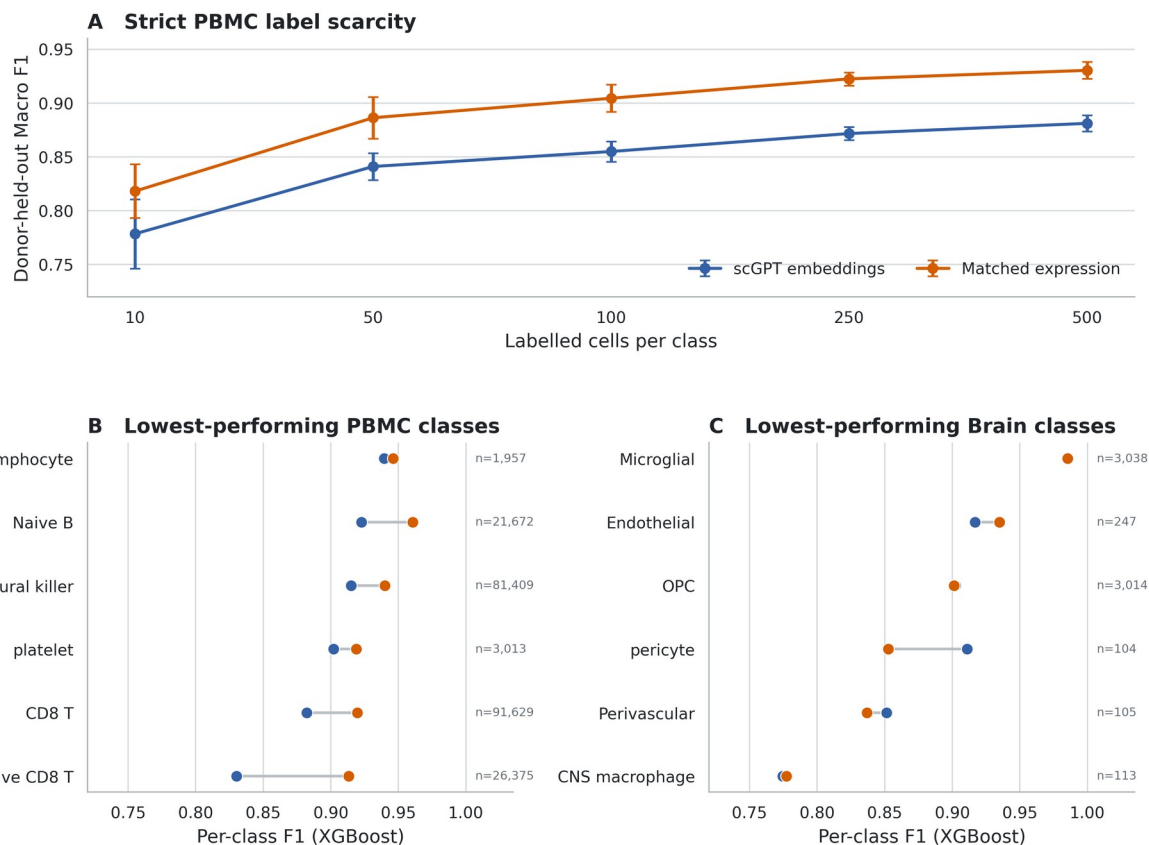


Figure 5. Strict scarcity and class-level performance. (A) Indonesia-PBMC Macro F1 when feature selection and model selection use only the indicated labelled cells. Points are means across five outer donor folds after averaging two sampling seeds within fold; bars are fold SD. (B–C) XGBoost F1 for the six lowest-performing portal classes by mean representation-level F1 in PBMC and brain.

Pig pancreas remains a cross-species stress test

The performance deficiency of the scGPT pipeline was most severe on the Pig Pancreas dataset (Macro F1 = 0.231), which is a cross-species dataset entirely outside scGPT's pre-training distribution. The relatively highest scGPT-pipeline performance was observed on TNBC Breast Cancer (Macro F1 = 0.783), a human oncology dataset, yet even here it was outperformed by all three classical methods (Logistic Regression: 0.996, XGBoost: 0.997, Random Forest: 0.995).

The porcine object contained one donor ID, so donor-held-out evaluation was impossible. Under the separate random-cell stress test, matched expression produced Macro F1 values of 0.988, 0.989 and 0.992 with Logistic Regression, Random Forest and XGBoost, respectively. Frozen

human-scGPT embeddings produced 0.966, 0.613 and 0.888, while SVD-512 produced 0.987, 0.938 and 0.967. These high values partly reflect interpolation among randomly separated cells from the same object. Because the whole-human checkpoint was applied through direct/uppercase symbol matching without an orthology map, these results are not included in the primary evidence and do not evaluate a principled cross-species scGPT workflow.

Discussion

The central finding of our study is unambiguous: when evaluated under realistic, out-of-the-box conditions on large-scale, diverse single-cell atlases, classical machine learning methods—particularly Logistic Regression and XGBoost on HVGs—substantially and consistently outperform the scGPT-embedding+LogReg frozen-representation linear-probing pipeline. This converges with a growing body of literature questioning the practical superiority of scLLMs over well-tuned classical pipelines. Kedzierska et al. (2025) reported that current scLLMs (scGPT, Geneformer, scFoundation, UCE) fail to consistently outperform simpler baselines in zero-shot transfer; Boiarsky et al. (2023) reached an analogous conclusion across multiple foundation models on cell-type classification, finding that classical learners often match or exceed scLLM-embeddings; and earlier benchmarking of cell-type classifiers by Abdelaal et al. (2019) likewise showed that simple linear methods are remarkably hard to beat on scRNA-seq label transfer. The strong performance of HVG-based linear classifiers is also consistent with the long-standing observation that scRNA-seq variation lies in a low intrinsic dimensionality (Heimberg et al. 2016), which is precisely the regime in which Logistic Regression on HVGs is near-optimal. Our work extends these prior reports in two important directions: (1) by explicitly benchmarking *statistical calibration* rather than accuracy alone—which, to our knowledge, has not been quantified in any of the above studies—and (2) by demonstrating that the performance gap holds

~~across six biologically distinct atlases spanning multiple tissues, species, and pathological contexts.~~

This benchmark addresses a deliberately narrow practical question: when cell labels are predicted from a fixed frozen-scGPT representation, how does the complete pipeline compare with supervised models trained on the same selected expression variables? The answer was atlas-dependent. Matched expression outperformed scGPT embeddings with every prespecified classifier in TNBC and PBMC, whereas scGPT embeddings were higher with all three classifiers in brain. Donor-cluster intervals supported all six TNBC and PBMC expression advantages, supported the scGPT advantage for brain Random Forest, and did not separate the brain Logistic Regression or XGBoost pipelines. The resulting conclusion is not that either representation is universally superior. It is that frozen scGPT utility depends on tissue, label structure and downstream learner, and that strong matched-expression models remain necessary controls.

~~Third, our results are based on a single stratified 80/20 train/test split per dataset, with HVG selection performed on the full dataset prior to the split. The split is a *random* stratified split that preserves class proportions but does *not* group by donor or batch; cells from the same donor may therefore appear in both the training and test folds, so the reported accuracies reflect within-cohort generalisation and may be optimistic relative to a donor- or batch-held-out protocol. Because all model families are evaluated on the identical split, this affects the absolute numbers rather than the cross-model ranking, but a donor-grouped split is an important refinement for future work. Test-set uncertainty is quantified by 1,000 bootstrap resamples (percentile method 95% CIs on Macro F1, reported throughout Tables and), which captures sampling variability for a fixed training set but not variability across alternative training partitions. Critically, because the same HVG set is provided to all four model families on each dataset, none of the cross-model~~

comparisons in this paper are biased by HVG leakage—only the absolute performance numbers may be modestly optimistic. The same dataset-wide HVG-selection convention has been adopted by prior benchmarks against which we compare (Abdelaal et al. 2019; Boiarsky et al. 2023; Kedzierska et al. 2025), so our numbers are directly comparable to that literature. We did not perform k -fold cross-validation: each of the six atlases requires a multi-hour scGPT embedding pass on a high-end GPU, and repeating the entire pipeline (including HVG selection nested within each fold) for all six datasets and all six fairness-control models, plus one end-to-end fine-tuning run, was outside our compute budget. A stricter nested k -fold CV protocol reporting mean $\pm$ std across folds is a worthwhile refinement for future work.

The donor-held-out design changes the meaning of the benchmark. Randomly separating cells within an atlas estimates interpolation when donor, sample and batch signatures can occur on both sides of the split. Grouping every donor's cells together instead asks whether a labelled training cohort supports annotation of an unseen donor processed within the same curated resource. The evaluated test sets therefore exclude direct donor overlap, but they do not constitute external-atlas validation. Acquisition protocols, annotation conventions, ancestry composition and study-specific curation remain shared within each resource. The appropriate interpretation is within-atlas unseen-donor performance, not broad biological generalization.

~~Fourth, our classical ML methods used default hyperparameters without optimisation, which may slightly underestimate their potential performance but reflects realistic out-of-the-box usage.~~

Fairness between representations requires more than supplying the same classifier name. The official scGPT embeddings are unit-normalized and 512-dimensional, while matched expression is sparse, differently scaled and approximately twice as wide. A fixed Logistic Regression penalty can act differently in these spaces. Independent searches over regularization and scaling

remove that avoidable asymmetry. The same principle applies to tree learners, whose feature-sampling and depth choices interact with representation geometry. Keeping the classifier fixed for each primary contrast also prevents a favourable head from being chosen after observing outer-test performance.

The SVD-512 control clarifies the role of compression. A training-fitted 512-component expression projection retained most of the uncompressed expression performance, with Logistic Regression differences of only 0.0005–0.0049 across the three atlases. SVD-512 preserved the matched-expression advantage over scGPT in TNBC and PBMC and the scGPT advantage in brain. Thus, the results are not consistent with a simple explanation in which reducing approximately 1,100 selected genes to 512 coordinates necessarily causes the observed losses. At the same time, SVD is not a substitute for scVI, does not model counts generatively and does not remove batch effects. Generative latent-variable baselines remain an important extension.

The brain result is an informative counterexample to a one-directional narrative. Rare macrophage, vascular and precursor classes contributed strongly to Macro F1 despite accounting for a small fraction of cells. scGPT may preserve features useful for separating some rare brain populations, or its pretraining distribution may be more favourable for this tissue. The present design cannot distinguish those mechanisms. Only 12 brain donors were available, and the conditional intervals were correspondingly wider than in TNBC and PBMC. The Random Forest interval nevertheless favoured scGPT, while Logistic Regression and XGBoost remained uncertain at the donor-resampling level. Reporting all three fixed heads prevents one of these outcomes from being elevated post hoc.

1. ~~**Data-scarce or novel biology settings.** In scenarios where labelled training data is genuinely unavailable and classical supervised methods are therefore not applicable,~~

frozen scLLM representations—when paired with appropriate uncertainty quantification—may provide a useful starting point, as suggested by prior reports that pre-trained scLLM embeddings transfer to small labelled cohorts and rare cell-type discovery (Yang et al. 2022; Theodoris et al. 2023; M. Hao et al. 2024), and as more broadly demonstrated by reference-mapping frameworks such as scArches (Lotfollahi et al. 2022), which leverage pre-trained representations to integrate small query datasets into existing reference atlases.

The strict scarcity experiment did not reveal a reversal in favour of frozen scGPT embeddings. Matched expression had higher Macro F1 at every tested label budget and in every outer fold even when the gene list was learned only from labelled cells. At the smallest budget, both representations exceeded 0.77 mean Macro F1 across 13 PBMC categories, showing that the supervised heads extracted useful signal from very limited labels. Calibration told a different story: scGPT embeddings sometimes gave lower uncalibrated ECE at larger budgets despite lower discrimination. This divergence reinforces that accuracy, calibration and representation utility are separate properties.

~~Second, the calibration analysis pertains to the end-to-end classification pipeline—scGPT embeddings followed by a downstream classifier—rather than to scGPT in isolation. The predicted probabilities used to compute the ECE originate from the downstream classifier, not from scGPT itself. Our defensible claim is therefore that the scGPT-embedding+LogReg pipeline is markedly less well-calibrated than classical HVG-based pipelines on these benchmarks, not that scGPT is intrinsically uncalibrated as a representation model.~~

~~Perhaps the most operationally consequential finding is the systematic miscalibration of the scGPT-linear-probe pipeline. Calibrated confidence scores are precisely what allow a practitioner~~

to triage predictions—trusting high-confidence calls and routing low-confidence ones to expert review. When a pipeline’s confidence scores do not track its empirical accuracy, as the scGPT-linear-probe pipeline’s do not (ECE up to 0.177), this triage breaks down: the confidence values can no longer be used as reliability estimates, regardless of the direction of the deviation. The phenomenon is well-characterised in deep learning more broadly: Guo et al. (2017) showed that modern overparameterised neural networks are frequently miscalibrated and proposed Expected Calibration Error and reliability diagrams as the standard diagnostic. Our results indicate that pipelines built on single-cell foundation model embeddings can inherit this miscalibration, and we argue that calibration should be measured and reported as routinely as accuracy (Ji et al. 2023).

Calibration must be attributed to the complete prediction pipeline. Frozen scGPT supplies a vector, not the class probability used here; the supervised head and its regularization shape confidence. Random Forest was substantially less calibrated than Logistic Regression and XGBoost in TNBC and PBMC for both representations. Temperature scaling improved mean ECE, Brier score and NLL in the TNBC and brain sensitivity analysis, but improvement was not universal at the individual fold level. A scalar temperature cannot repair class confusion or missing biological information, and improved ECE does not guarantee simultaneous improvement in every proper score. Reliability-bin counts, Brier score and NLL should therefore accompany ECE, preferably under external cohort shift.

Class-level results similarly limit any single-number account. In TNBC, the main difference was not a general failure of one representation but a concentration of errors at B-lineage, T-cell and myeloid boundaries. In PBMC, naive and conventional CD8 T cells overlapped with CD4 T and natural-killer categories, with larger errors from scGPT embeddings. Brain errors clustered

among biologically related or rare populations, including central-nervous-system macrophages versus microglia and vascular-associated cells versus oligodendrocytes. Portal categories provide reproducible targets, but they are not an error-free biological ground truth. Differences may reflect transcriptional continuity, donor-specific abundance, annotation granularity or label harmonization as well as representation quality.

Cross-species gene handling (Pig Pancreas):

The pig analysis illustrates why dataset count should not be confused with independent evidence. One donor value cannot support donor-held-out inference, and direct uppercase symbol matching is not a defensible orthology workflow. The high random-split scores show only that the pipelines can interpolate broad categories within this object. A dedicated cross-species evaluation would require an explicit orthologue map, species-aware preprocessing, multiple animals and animal-held-out validation.

The uncertainty analysis improves the biological unit of inference but remains conditional. Donor-cluster resampling preserves within-donor dependence and pairs the representation contrast. It does not refit HVG selection, SVD or models. Repeating the complete nested workflow inside each bootstrap replicate would quantify additional training and selection variability but would be substantially more computationally demanding. The reported intervals therefore show sensitivity to the observed donor composition after fitting. Independent cohorts remain the strongest test of reproducibility.

~~First, we evaluated only one scLLM (scGPT). The frozen representation pipeline was tested with three downstream learners (Section), and an additional end-to-end fine-tuning experiment was performed on TNBC (Section); however, alternative scLLMs (Geneformer, scFoundation, UCE) may show different accuracy and calibration profiles, and the end-to-end fine-tuning experiment~~

was performed on a single dataset only. We do not generalise our results beyond the scGPT family of models.

Fifth, the GRN analysis is an embedding-derived gene co-expression similarity analysis rather than an attention-based or causal regulatory inference, and it is qualitative: the recovered modules are validated against the published immunology literature for each module (Section ; (Reith and Mach 2001; Voskoboinik, Whisstock, and Trapani 2015; Smith-Garvin, Koretzky, and Jordan 2009)), but we did not quantify enrichment against a null or baseline graph (random gene sets, a PCA-derived similarity graph, or a raw-expression correlation graph), and experimental in-vitro validation of any putatively novel predictions remains beyond the scope of this benchmarking study. A quantitative, null-anchored network benchmark is a natural extension.

Sixth, our benchmark compares scGPT against generic machine-learning baselines (Logistic Regression, Random Forest, XGBoost) but does not include established reference-based or marker-based single-cell annotation tools such as SingleR (Aran et al. 2019), Azimuth (Y. Hao et al. 2021), scType (Ianevski, Giri, and Aittokallio 2022), scPred (Alquicira-Hernandez et al. 2019), CellTypist (Domínguez Conde et al. 2022), or scmap (Kiselev, Yiu, and Hemberg 2018). Comprehensive head-to-head comparisons among such tools have been reported elsewhere (Abdelaal et al. 2019; Pasquini et al. 2021); our contribution is specifically the head-to-head comparison of an scLLM against the supervised learning baseline that scLLMs are most often implicitly positioned to replace. A direct benchmark of scGPT against the reference-based annotators above (using the same six atlases, the same evaluation protocol, and the same calibration metrics) is a natural and important follow-up, and we expect it to further sharpen the cost-benefit picture for clinical deployment.

Seventh, our calibration audit reports ECE and the bin-level reliability diagrams but not the scalar Brier score, negative log-likelihood, or signed and class-wise ECE decompositions, because per-cell predicted probability arrays were not retained from the original runs. Since ECE is an unsigned summary and those arrays would be required to compute signed or strictly proper scoring rules, we restrict our claims to the *magnitude* of miscalibration captured by ECE and the reliability diagrams, and do not interpret its direction.

Several limitations define the evidence boundary. The human benchmark contains three atlases and does not cover the full diversity of tissues, technologies or disease contexts; in particular, a completed donor-held-out TME or human-pancreas analysis is not part of the evidence. SVD-512 Random Forest and XGBoost were not available as complete five-fold pipelines for PBMC and were omitted rather than imputed. Temperature scaling was restricted to TNBC and brain. Only scGPT 0.2.5, one released whole-human checkpoint and one frozen embedding interface were studied. Official end-to-end fine-tuning, newer checkpoints, scVI-like baselines and other foundation models were not tested. Public checkpoint pretraining may overlap source studies, and available metadata do not support a complete cell-level overlap audit. The target labels are broad and differ in granularity among atlases. Finally, the benchmark concerns supervised annotation of known categories rather than novel-state discovery, perturbation response, regulatory inference or multimodal transfer.

Within those boundaries, the study provides a reproducible evaluation template. Foundation-model embeddings should be compared with strong matched inputs, tuned on training groups, assessed under biological holdout and evaluated as complete representation-classifier pipelines. Compression controls and class-level errors are needed to interpret performance differences, while calibration should be reported with multiple probability metrics and a clearly identified

calibration cohort. These practices remain useful whether a pretrained representation wins, ties or loses.

Conclusions

~~We present, to our knowledge, a focused benchmarking of statistical calibration in pipelines built on scGPT at scale. Across six large, diverse single-cell atlases totalling 1.3 million cells, classical machine learning methods—XGBoost, Logistic Regression, and Random Forest trained on highly variable genes—substantially and consistently outperform a frozen representation linear probing pipeline built on scGPT in both predictive accuracy (Macro F1) and calibration (ECE). Critically, the scGPT-embedding+LogReg pipeline is systematically miscalibrated across all evaluated datasets, with ECE far exceeding that of every classical baseline, so that its confidence scores cannot be used as reliability estimates. Despite this, scGPT embeddings retain biologically meaningful structure, as evidenced by the recapitulation of known co-expression relationships in zero-shot, embedding-derived gene network inference.~~

~~Our findings argue for three immediate changes in how the field evaluates and deploys scLLMs: (1) calibration metrics should be mandatory in all scLLM benchmarking publications, with a clear distinction between calibration of the foundation model and calibration of the downstream pipeline; (2) classical ML baselines should be rigorously benchmarked against scLLM pipelines—using matched downstream learners and matched input gene sets—before claiming scLLM superiority; and (3) the practical and financial costs of scLLM deployment must be weighed against demonstrated performance advantages. Until scLLMs can demonstrate calibrated, reproducible superiority over classical methods under matched and externally validated~~

~~benchmark conditions, their routine adoption for cell-type annotation in large-scale atlases is premature.~~

Pretrained representations are often assumed to provide a more powerful starting point for cell annotation than conventional expression features. This study shows why that assumption must be tested rather than accepted. Carefully matched expression models were more accurate in the TNBC and PBMC atlases, whereas frozen scGPT representations were more useful in brain. The practical lesson is that the best representation depends on the tissue, the label structure and the intended use; it cannot be chosen from the reputation or scale of the underlying model alone.

For researchers applying single-cell foundation models, frozen scGPT embeddings should therefore be treated as one candidate representation, not as a default replacement for transparent expression baselines. Biological donor holdout, fair model tuning, confidence quality and class-specific errors all matter when deciding whether a pretrained representation adds value. The broader contribution of this benchmark is a practical framework for making that decision. It identifies settings in which frozen scGPT helps, settings in which simpler models remain stronger, and the evidence needed before claiming an advantage. Whether official fine-tuning, newer models or transfer across independent cohorts changes this balance remains an important question for future work.

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Declarations

Ethics approval and consent to participate. Not applicable. This study used only publicly available, fully de-identified single-cell RNA-sequencing datasets distributed through the Chan Zuckerberg CELL×GENE Discover portal under the original data generators’ ethics approvals. No new participants were recruited and no new specimens were collected.

Consent for publication. Not applicable.

Availability of data and materials. The four datasets reported in this study are publicly available through the CZ CELL×GENE Discover portal. For each dataset, the collection page, interactive Explorer page and direct H5AD download are provided below.

TNBC breast cancer. Collection: <https://cellxgene.cziscience.com/collections/ceef2841-5333-46ac-92ef-ccbe0c20fe55>; Explorer: <https://cellxgene.cziscience.com/e/6f9de485-58cd-4342-bfc4-b3d3dd223aa8.cxg/>; Direct H5AD: <https://datasets.cellxgene.cziscience.com/af8c4fce-4c63-4671-b339-91a383cf36f6.h5ad>

Indonesia PBMC. Collection: <https://cellxgene.cziscience.com/collections/d1e0e64d-6d2a-4a3e-b7f4-43ed909a9d9c>; Explorer: <https://cellxgene.cziscience.com/e/ca7d95ac-53aa-4a94-8c76-300c608ce6a0.cxg/>; Direct H5AD: <https://datasets.cellxgene.cziscience.com/665714af-4be5-49a3-913b-5ab5ac25620d.h5ad>

Brain atlas. Collection: <https://cellxgene.cziscience.com/collections/0986e4cd-7a58-405d-9b91-4b199bb4124e>; Explorer: <https://cellxgene.cziscience.com/e/e2a00644-0a48-4815-ae87-d562045114f5.cxg/>; Direct H5AD: <https://datasets.cellxgene.cziscience.com/0ab54d91-066c-4223-a9ea-6a3b0d1adef4.h5ad>

Pig pancreas. Collection: <https://cellxgene.cziscience.com/collections/0a77d4c0-d5d0-40f0-aa1a-5e1429bcbd7e>; Explorer: <https://cellxgene.cziscience.com/e/db4a9ed2-e994-40c1-b7ec-4091fdf7b6c1.cxg/>; Direct H5AD: <https://datasets.cellxgene.cziscience.com/55cfae87-6348-44df-a4ed-c132569dea54.h5ad>

The released whole-human scGPT checkpoint and source are available from <https://github.com/bowang-lab/scGPT>. Analysis code and aggregate outputs are available under the MIT licence at <https://github.com/alirezakhosravii/scGPTvsClassical>.

Competing interests. The authors declare that they have no competing interests.

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